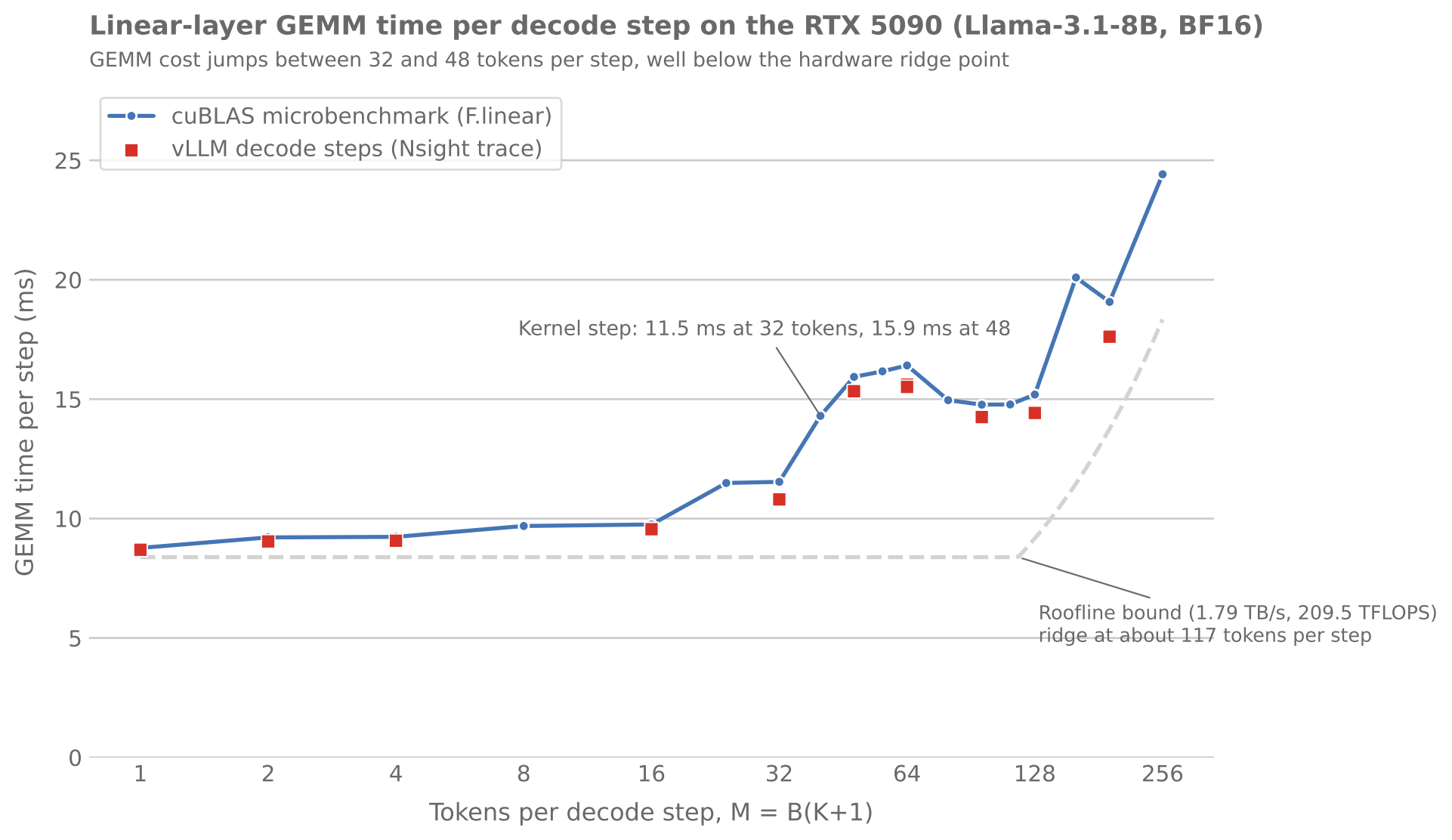
Linear-layer GEMM time per decode step on the RTX 5090 (Llama-3.1-8B, BF16)
GEMM cost jumps between 32 and 48 tokens per step, well below the hardware ridge point
cuBLAS microbenchmark (F.linear)
vLLM decode steps (Nsight trace)
GEMM time per step (ms)
0
5
10
15
20
25
1
2
4
8
16
32
64
128
256
Tokens per decode step, M = B(K+1)
Kernel step: 11.5 ms at 32 tokens, 15.9 ms at 48
Roofline bound (1.79 TB/s, 209.5 TFLOPS)
ridge at about 117 tokens per step